That is 25 percent of data will lie below $Q_1$, 50 percent of data below $Q_2$ and 75 percent below $Q_3$. Here $Q_2$ is called the Median. Quartiles are obtained in almost the same way as median

## Quartiles for Raw or Ungrouped data:

If the data set consist of n items and arranged in ascending order then

$$Q_1 = \left(\frac{n+1}{4}\right)^{th} \text{ item}, \quad Q_2 = \left(\frac{n+1}{2}\right)^{th} \text{ item} \quad \text{and} \quad Q_3 = 3\left(\frac{n+1}{4}\right)^{th} \text{ item}$$

### Example 5.30

Compute $Q_1$ and $Q_3$ for the data relating to the marks of 8 students in an examination given below 25, 48, 32, 52, 21, 64, 29, 57

***Solution:***

$$n = 8$$

Arrange the values in ascending order

21, 25, 29, 32, 48, 52, 57, 64 we have

$$Q_1 = \left(\frac{n+1}{4}\right)^{th} \text{ item}$$

$$= \left(\frac{8+1}{4}\right)^{th} \text{ item}$$

$$= 2.25^{\text{th}} \text{ item}$$

$$= 2^{\text{nd}} \text{ item} + \left(\frac{1}{4}\right) \ (3^{\text{rd}} \text{ item} - 2^{\text{nd}} \text{ item})$$

$$= 25 + 0.25\ (29 - 25)$$

$$= 25 + 1.0$$

$$Q_1 = 26$$

$$Q_3 = 3\left(\frac{n+1}{4}\right)^{th} \text{ item}$$

$$= 3 \times (2.25)^{th} \text{ item}$$

$$= 6.75^{\text{th}} \text{ item}$$

$$= 6^{\text{th}} \text{ item} + 0.75 \ (7^{\text{th}} \text{ item} - 6^{\text{th}} \text{ item})$$



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